

Continuous 3-D Latent Diffusion for Medical Generation and Reconstruction

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Abstract—High-resolution three-dimensional (3-D) medical diffusion models remain constrained by the cost of processing full volumes, even when denoising is performed in a compact latent space. We introduce a continuous 3-D latent diffusion model (LDM) framework for computed tomography (CT) and magnetic resonance imaging (MRI) generation and measurement-guided reconstruction. Its central component is a compact autoencoder (AE) with a coordinate-conditioned local implicit image function (LIIF) decoder that represents a volume as a continuous function of spatial coordinates. By evaluating the convolutional decoder once on the latent grid and restricting repeated computation to a lightweight implicit head, the proposed design avoids overlapping sub-volume decoding while remaining differentiable for inverse-problem optimization. We evaluate the framework on CT volumes of 512^3 voxels and MRI volumes of 256^3 voxels. On high-resolution CT, the proposed AE is approximately $\times 12$ – $\times 32$ faster than the evaluated reference autoencoders, achieves the lowest peak graphics processing unit (GPU) memory use, and retains comparable structural fidelity despite a moderate reduction in voxel-level accuracy. The resulting frozen 3-D latent prior generates coherent full volumes without visible patch seams and can be applied, without task-specific retraining, to sparse-view CT and accelerated MRI reconstruction through hard data consistency. Although direct pixel-domain reconstruction remains more accurate, the results demonstrate that a single volumetric latent prior can support both unconditional generation and measurement-conditioned reconstruction on one GPU. Overall, the framework provides a practical trade-off between continuous volumetric decoding, computational efficiency, and fine-detail preservation.

Our code will be made available at <https://github.com/mellak/>.

Index Terms—3-D medical image generation, computed tomography (CT), magnetic resonance imaging (MRI), latent diffusion model (LDM), local implicit image function (LIIF), implicit neural representation (INR), image reconstruction.

I. INTRODUCTION

High-resolution medical images such as CT and MRI are inherently 3-D: anatomical structures extend across slices, and image quality must remain consistent in the axial, coronal, and sagittal planes. Volumetric coherence is therefore essential for both image generation and reconstruction. Diffusion models (DMs) have become a reference approach for image generation [1] and have also been applied to native 3-D medical image synthesis [2]. However, clinical volumes

commonly reach 512^3 voxels or larger, making direct voxel-space modeling computationally demanding. At each denoising step, a voxel-space DM evaluates a dense 3-D convolutional neural network (CNN) over the entire volume and stores large intermediate activation maps. A practical framework must therefore preserve global anatomical coherence without repeatedly processing every voxel at full resolution.

LDMs reduce this cost by moving the diffusion process from image space to a compact latent representation. An AE first compresses the input volume, after which the generative model is trained on the resulting latent codes rather than directly on voxels [3]. This strategy has enabled the modeling of anatomical variability in compressed 3-D brain MRI representations [4]. Subsequent methods have pursued different strategies for scaling volumetric generation. PatchDDM trains DMs on local 3-D patches and applies them to larger volumes at inference [5]. MedSyn and LAND incorporate anatomical conditioning for high-resolution chest CT synthesis [6], [7]. More recent frameworks, including MAISI, MAISI-v2, and 3D MedDiffusion, combine latent compression, anatomical conditioning, and patch-based processing to generate CT and MRI volumes at resolutions up to 512^3 [8]–[10].

Despite reducing the cost of the generative model, LDMs do not eliminate the computational burden associated with high-resolution volumes. Instead, part of this burden remains in the AE. The encoder must map a full-resolution volume to the latent space, while the decoder must reconstruct the latent representation on the original voxel grid. At 512^3 resolution, a dense encoding or decoding operation may exceed the memory capacity of a single GPU. This limitation becomes particularly restrictive when the decoder is used repeatedly within an iterative reconstruction algorithm.

Existing AEs generally address this limitation through spatial decomposition. Hierarchical and sub-volume methods combine coarse global representations with high-resolution local regions [11]. Volume-compression and tensor-splitting strategies reduce the cost of dense 3-D encoding and decoding [8], while patch-volume AEs process the input and output as multiple smaller regions [10]. Related approaches partition the generative model itself through patch-based diffusion or position-aware score blending [12], [13]. Although these methods reduce peak memory, they generally require repeated evaluation of a dense network on multiple, often overlapping, sub-volumes, followed by aggregation or blending. Their runtime therefore increases with the number of processed regions, which is especially costly when decoding must be performed repeatedly.

Coordinate-based decoding provides an alternative to dense,

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fixed-grid volume reconstruction [14]–[17]. Implicit neural representations (INRs) represent a signal as a function of spatial coordinates, making them resolution-agnostic, differentiable, and queryable at arbitrary locations [18]–[21]. This principle was initially developed mainly for two-dimensional (2-D) image representation: LIIF conditions coordinate queries on local latent features, thereby connecting a discrete feature grid to a continuous image representation [22], while implicit decoders have been combined with LDMs to enable arbitrary-scale generation without moving the diffusion process out of latent space [23]. In medical imaging, coordinate-based models have subsequently been coupled with measurement operators for reconstruction from incomplete observations [21], [24] and combined with physical models and generative priors to enforce data consistency in ill-posed inverse problems [25], [26]. For the present work, two properties are particularly important: the decoder can render an image through coordinate queries rather than dense sub-volume processing, and the resulting mapping remains differentiable with respect to the latent representation.

The development of continuous representations has recently extended from 2-D images and scan-specific optimization to learned volumetric medical models. In particular, MedIL introduced an implicit latent representation for heterogeneous-resolution brain MRI and lung CT, enabling learned 3-D generation at arbitrary output resolutions [27]. This work demonstrates the value of continuous decoding for accommodating heterogeneous image dimensions and voxel spacings. However, the use of such a decoder for efficient full-volume processing at 512^3 resolution, and especially for repeated differentiation within a measurement-guided reconstruction procedure, remains largely unexplored. The question addressed in this work is therefore whether a coordinate-conditioned AE can process full high-resolution medical volumes on a single GPU while remaining sufficiently efficient and differentiable to serve as an interface between a learned 3-D prior and a medical acquisition operator.

To address this gap, we propose LIIF-AE, a 3-D AE in which a lightweight encoder maps a full-resolution volume to a compact latent grid, while a coordinate-conditioned LIIF decoder represents voxel intensities as a continuous function of spatial coordinates. Our contributions are threefold. First, we design a memory-efficient encoder that performs dense, wide-channel processing only after spatial downsampling, enabling full 512^3 volume encoding and decoding on a single GPU. Second, the proposed decoder applies its convolutional stage once to the latent grid and renders the volume through memory-bounded coordinate queries, avoiding overlapping sub-volume decoding while preserving end-to-end differentiability. Third, we integrate this representation with a LDM and demonstrate its use for both unconditional volume generation and measurement-guided reconstruction. We generate full-volume CT and MRI images at 512^3 and 256^3 resolution, respectively, and reuse the same frozen latent prior for sparse-view CT and undersampled MRI reconstruction within the diffusion-based posterior-sampling framework [28], using the hard-data-consistency procedure of ReSample [29].

The remainder of this paper is organized as follows. Sec-

tion II presents the proposed framework, Section III describes the experiments and results, Section IV discusses the findings and limitations, and Section V concludes the paper.

II. MATERIALS AND METHODS

The hyperparameter values in the proposed methodology are provided in the source code (address provided in abstract).

A. Problem Formulation and Overview

Let $\mathbf{x} \in \mathcal{X} \triangleq \mathbb{R}^{D \times H \times W}$ denote a high-resolution 3-D image volume, where D, H, W are the spatial dimensions and $[\mathbf{x}]_i$ denotes the image intensity at voxel i . We address two tasks under constrained memory. The generation problem is to learn the data distribution $p(\mathbf{x})$, from which new realistic volumes can be sampled. The reconstruction problem is to recover $\mathbf{x}$ from incomplete measurements $\mathbf{y} \in \mathcal{Y} \triangleq \mathbb{R}^n$ acquired through the forward model

$$\mathbf{y} = \mathcal{A}(\mathbf{x}) + \boldsymbol{\eta}, \quad (1)$$

where $\mathcal{A}: \mathcal{X} \rightarrow \mathcal{Y}$ is an acquisition operator, instantiated in this work as a projection operator for CT or an undersampled Fourier transform for MRI, and $\boldsymbol{\eta}$ denotes a centered measurement noise. The combined problem is therefore to estimate a plausible volume under the learned prior $p(\mathbf{x})$ that is consistent with the measurements, i.e., to find $\mathbf{x}$ such that

$$\text{find } \mathbf{x} \sim p(\mathbf{x}) \text{ s.t. } \mathbf{y} \approx \mathcal{A}(\mathbf{x}). \quad (2)$$

Solving (2) is equivalent to sampling from the posterior $\mathbf{x} \sim p(\mathbf{x} | \mathbf{y}) \propto p(\mathbf{y} | \mathbf{x}) p(\mathbf{x})$, where the likelihood $p(\mathbf{y} | \mathbf{x})$ is defined by the forward model (1). In this work, the prior $p(\mathbf{x})$ is learned using a DM [1], and we used a diffusion posterior sampling (DPS) algorithm inspired by Zhu *et al.* [30] and Song *et al.* [29].

To make both tasks tractable at clinical resolution on a single GPU, $\mathbf{x}$ is represented by a compact latent variable $\mathbf{z} \in \mathcal{Z} \triangleq \mathbb{R}^{C_z \times d \times h \times w}$, with latent channels C_z and latent spatial dimensions $(d, h, w) = (D/r, H/r, W/r)$ obtained from a spatial downsampling factor r ; the values of C_z and r are specified in Section II-B1. The latent is produced and decoded by a jointly trained encoder–decoder pair $(\mathcal{E}_\phi, \mathcal{D}_\psi)$ with trainable parameters ϕ and ψ . The encoder $\mathcal{E}_\phi$ has two components: the encoder mean $\boldsymbol{\mu}_\phi: \mathcal{X} \rightarrow \mathcal{Z}$ and the encoder standard deviation (STD) $\boldsymbol{\sigma}_\phi: \mathcal{X} \rightarrow \mathcal{Z}$, which are used to sample a latent variable $\mathbf{z}$ from an image $\mathbf{x}$, i.e.,

$$\begin{aligned} q_\phi(\mathbf{z} | \mathbf{x}) &= \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_\phi(\mathbf{x}), \boldsymbol{\Sigma}_\phi(\mathbf{x})), \\ \mathbf{z} &\sim q_\phi(\mathbf{z} | \mathbf{x}), \\ \boldsymbol{\Sigma}_\phi(\mathbf{x}) &= \text{diag}(\boldsymbol{\sigma}_\phi^2(\mathbf{x})), \\ (\boldsymbol{\mu}_\phi(\mathbf{x}), \boldsymbol{\sigma}_\phi(\mathbf{x})) &= \mathcal{E}_\phi(\mathbf{x}). \end{aligned} \quad (3)$$

The inverse problem (2) can then be reformulated in the latent space by replacing the image prior $p(\mathbf{x})$ with a prior $p(\mathbf{z})$ over latent representations. Specifically, we solve

$$\text{find } \mathbf{z} \sim p(\mathbf{z}) \text{ s.t. } \mathbf{y} \approx \mathcal{A} \circ \mathcal{D}_\psi(\mathbf{z}). \quad (4)$$

Generation and reconstruction are then both posed on the latent space $\mathcal{Z}$ through a single diffusion prior $p(\mathbf{z})$ trained

on encoded latents, so that $p(\mathbf{x})$, the model's distribution over volumes, is the pushforward of $p(\mathbf{z})$ through $\mathcal{D}_\psi$.

In this work, the decoder $\mathcal{D}_\psi$ is implicit: it predicts intensities at queried spatial coordinates and can therefore render a volume, or a coordinate subset, in memory-bounded chunks rather than instantiating the full $D \times H \times W$ grid at once.

The framework follows a two-stage design. The AE pair $(\mathcal{E}_\phi, \mathcal{D}_\psi)$ is trained first and then frozen (Section II-B); the diffusion prior is trained on the resulting latents (Section II-C) and reused, without modification, as a prior that regularizes reconstruction from $\mathbf{y}$ via decoded data consistency $\mathcal{A} \circ \mathcal{D}_\psi$ (Section III-D). The implicit decoder is evaluated in coordinate chunks, allowing the composition $\mathcal{A} \circ \mathcal{D}_\psi$ to be differentiated with bounded memory.

B. Latent Autoencoder

In the following, for notational conciseness, the dependence on (ϕ, ψ) is made explicit through $\mathcal{E}_\phi$ and $\mathcal{D}_\psi$. Consequently, the dependence of the training objectives in Section II-B3 on (ϕ, ψ) is expressed through the corresponding encoded and decoded outputs.

1) *Encoder*: The encoder $\mathcal{E}_\phi$ operates in two stages, i.e.,

$$\mathcal{E}_\phi = \mathcal{E}_{\phi_2}^{\text{enc}} \circ \mathcal{E}_{\phi_1}^{\text{down}}, \quad \phi = (\phi_1, \phi_2). \quad (5)$$

The first stage, $\mathcal{E}_{\phi_1}^{\text{down}}$, reduces the spatial resolution by a factor of $r = 4$ along each axis using a small stack of lightweight, depthwise-separable convolutional blocks interleaved with two strided convolutions:

$$\mathbf{h} = \mathcal{E}_{\phi_1}^{\text{down}}(\mathbf{x}) \in \mathbb{R}^{C_h \times d \times h \times w}, \quad (6)$$

where C_h is the hidden channel width. Depthwise convolutions mix space and channels separately, so their cost grows linearly, rather than quadratically, with the channel count. This lets $\mathcal{E}_{\phi_1}^{\text{down}}$ stay cheap in activation memory even though it operates on the largest, full-resolution feature maps.

The second stage, $\mathcal{E}_{\phi_2}^{\text{enc}}$, applies standard convolutional residual blocks at full channel width on the already-reduced grid $\mathbf{h}$, which contains 64 fewer voxels than $\mathbf{x}$. Deferring dense, wide-channel computation until after this reduction is what makes it affordable: $\mathcal{E}_{\phi_2}^{\text{enc}}$ can spend most of the encoder's capacity exactly where the posterior is read from, without inflating peak activation memory. It defines the encoder mean and the encoder STD,

$$(\boldsymbol{\mu}_\phi(\mathbf{x}), \boldsymbol{\sigma}_\phi(\mathbf{x})) = \mathcal{E}_{\phi_2}^{\text{enc}}(\mathbf{h}) \in \mathbb{R}^{2 \times C_z \times d \times h \times w}, \quad (7)$$

which define the factorized Gaussian posterior $q_\phi(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_\phi, \boldsymbol{\Sigma}_\phi)$, with $\boldsymbol{\Sigma}_\phi$ defined (3)¹ We use $C_z = 4$.

2) *Implicit Decoder*: We regard the discrete image $\mathbf{x} \in \mathcal{X} = \mathbb{R}^{D \times H \times W}$ as the sampling of a continuous image on the normalized domain $\Omega \triangleq [-1, 1]^3$, writing, with a slight abuse of notation, $\mathbf{x}(\mathbf{c}) \in \mathbb{R}$ and $\hat{\mathbf{x}}(\mathbf{c}) \in \mathbb{R}$ for the true and estimated intensities at a location $\mathbf{c} \in \Omega$. The voxels of $\mathbf{x}$ correspond to the uniform grid $\mathcal{G} = \{\mathbf{c}_i \in \Omega \mid i = 1, \dots, DHW\}$, which, the underlying image being continuous, may in principle be

chosen at any resolution, while the latent representation $\mathbf{z}$ is defined on a coarse grid $\tilde{\mathcal{G}}$ of size $d \times h \times w$, obtained by downsampling the image grid by a factor r along each spatial axis. In this view, decoding is no longer the construction of a dense $D \times H \times W$ tensor, which is what makes conventional decoders expensive, but the evaluation of the continuous image at any queried $\mathbf{c}$, and it decomposes into two operations: describing the local image content at each cell of $\tilde{\mathcal{G}}$, and combining the corresponding predictions at the query point. We factor the decoder into a feature-decoding stage and a coordinate-decoding stage,

$$\mathcal{D}_\psi = \mathcal{D}_{\psi_2}^{\text{mlp}} \circ \mathcal{D}_{\psi_1}^{\text{feat}}, \quad \psi = (\psi_1, \psi_2). \quad (8)$$

The first stage maps the compact latent grid to a richer feature grid. Since the diffusion model operates on $\mathbf{z}$, the number of latent channels C_z is kept small. A stack of stride-one convolutional residual blocks therefore expands the channel dimension while preserving the latent-grid resolution,

$$\mathbf{u} = \mathcal{D}_{\psi_1}^{\text{feat}}(\mathbf{z}) \in \mathbb{R}^{C_u \times d \times h \times w}, \quad C_u > C_z. \quad (9)$$

Because these convolutions are applied on the coarse grid $\tilde{\mathcal{G}}$ rather than the full image grid $\mathcal{G}$, this stage remains memory efficient.

The second stage renders voxel intensities at arbitrary coordinates using the LIIF local ensemble. The vector-valued decoder $\mathcal{D}_{\psi_2}^{\text{mlp}}$ is defined using a shared real-valued multi-layer perceptron (MLP), $\mathcal{D}_{\psi_2}^{\text{mlp}}$. For each query coordinate $\mathbf{c}_i$, the MLP predicts an intensity from each neighboring latent feature, and these predictions are combined using normalized trilinear weights:

$$\hat{x}(\mathbf{c}_i) = \sum_{j \in \mathcal{N}(\mathbf{c}_i)} w_{ij} \mathcal{D}_{\psi_2}^{\text{mlp}}(\mathbf{u}_j, \Delta \mathbf{c}_{ij}, v_i), \quad i = 1, \dots, DHW. \quad (10)$$

Here, $\mathcal{N}(\mathbf{c}_i)$ is the set of eight latent-grid cells neighboring $\mathbf{c}_i$, $\mathbf{u}_j \in \mathbb{R}^{C_u}$ is the feature vector at the center $\tilde{\mathbf{c}}_j$ of the j -th cell, and $\Delta \mathbf{c}_{ij} = \mathbf{c}_i - \tilde{\mathbf{c}}_j$ is the relative offset between the query coordinate and that cell center. Moreover, v_i denotes the query-cell size, and w_{ij} is the normalized trilinear weight satisfying $\sum_{j \in \mathcal{N}(\mathbf{c}_i)} w_{ij} = 1$. This formulation enables continuous intensity prediction on arbitrary query grids.

During training, we sample $N \ll DHW$ coordinates per iteration. During inference, the full volume is rendered in memory-bounded query chunks. Thus, decoding requires no overlapping sub-volumes or blending, remains differentiable with respect to $\mathbf{z}$, and can be used inside the reconstruction procedure of Section III-D.

3) *Training Objective*: We train the AE with a regularized reconstruction objective. At each iteration, we sample N coordinates from the target volume and minimize

$$\begin{aligned} \mathcal{L} = & \frac{1}{N} \sum_{i=1}^N |\hat{x}(\mathbf{c}_i) - x(\mathbf{c}_i)| \\ & + \lambda_{\text{KL}} D_{\text{KL}}(q_\phi(\cdot \mid \mathbf{x}) \parallel \mathcal{N}(\mathbf{0}_Z, \mathbf{I}_Z)). \end{aligned} \quad (11)$$

where $\hat{x}(\mathbf{c}_i)$ is given by (10) and $q_\phi(\cdot \mid \mathbf{x})$ by (7).

¹In our implementation, the encoder STD was implemented as $\boldsymbol{\sigma}_\phi = \exp(\mathbf{s}_\phi)$ where $\mathbf{s}_\phi$ is the actual second output component of $\mathcal{E}_\phi$.

C. Latent Diffusion Model

With $\mathcal{E}_\phi$ and $\mathcal{D}_\psi$ fixed, a 3-D U-Net noise predictor ϵ_θ parametrized by θ is trained as an unconditional denoising diffusion probabilistic model (DDPM) directly on latents $\mathbf{z} = \mu_\phi(\mathbf{x})$ from the training set,

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{\mathbf{z}, \epsilon, t} [\|\epsilon - \epsilon_\theta(\mathbf{z}_t, t)\|_2^2], \quad (12)$$

where $\epsilon \sim \mathcal{N}(\mathbf{0}_Z, \mathbf{I}_Z)$, $\mathbf{z}_t$ is a degraded version of $\mathbf{z}$ at uniformly sampled time step $t = 1, \dots, T$ and noise schedule $\{\alpha_t\}_{t=1}^T$, i.e.,

$$\mathbf{z}_t = \sqrt{\alpha_t} \mathbf{z} + \sqrt{1 - \alpha_t} \epsilon \quad (13)$$

with $\bar{\alpha}_t = \prod_{t'=1}^t \alpha_{t'}$

In this setting, the DM operates in the lower-dimensional latent space $\mathcal{Z}$ rather than on the image space $\mathcal{X}$. The trained noise predictor ϵ_θ defines an implicit generative prior $p(\mathbf{z})$ over $\mathcal{Z}$, reused by both downstream tasks below.

D. Measurement-Guided Reconstruction

For reconstruction, we reuse the frozen latent prior $p(\mathbf{z})$ and solve the latent inverse problem (4) using the hard-data-consistency procedure of ReSample [29], extended from 2-D images to 3-D latent volumes.

At each reverse step, Tweedie's formula provides an estimate $\hat{\mathbf{z}}_{0|t}$ of the clean latent representation. The estimate is then corrected toward the measurements $\mathbf{y}$ using three stages.

At high noise levels, $t > t_1$, no correction is applied because the decoded estimate is not yet sufficiently reliable.

At intermediate noise levels, $t_2 < t \leq t_1$, data consistency is enforced in image space. The decoded volume $\mathcal{D}_\psi(\hat{\mathbf{z}}_{0|t})$ initializes a regularized least-squares problem in $\mathbf{x}$, which is solved using conjugate gradients. This stage is computationally efficient because the decoder output is treated as fixed and no backpropagation through the decoder is required. The corrected volume is then re-encoded into the latent space using the posterior mean $\mu_\phi(\hat{\mathbf{x}}_0)$. However, latent autoencoders are lossy and only approximately invertible [31]. Consequently, re-encoding the corrected image may discard part of the data-consistency update and smooth fine details.

At low noise levels, $t \leq t_2$, data consistency is therefore enforced directly in latent space. Starting from $\hat{\mathbf{z}}_{0|t}$, the latent representation is optimized through $\mathcal{A} \circ \mathcal{D}_\psi$. This stage is more expensive because it requires backpropagation through the decoder, but it refines fine structures and reduces errors introduced by the image-space correction and re-encoding step [29, App. B]. The implicit decoder is evaluated in coordinate chunks to keep the memory cost bounded.

After each data-consistency correction, the corrected latent estimate is mapped back to the diffusion trajectory using the StochasticResample step of ReSample [29, Prop. 2]. The complete three-stage reconstruction procedure is summarized in Algorithm 1.

For reconstruction, we follow the medical-image settings of ReSample [29]. We use $T = 1000$ denoising diffusion implicit models (DDIM) steps and divide the reverse process into three intervals: $t > 750$, $300 < t \leq 750$, and $t \leq 300$. Image-space data consistency is solved using 50 conjugate-gradient iterations with relaxation $\kappa = 0.9$, while latent-space

Algorithm 1 Latent-Implicit 3-D Reconstruction with ReSample [29]

Require: $\mathbf{y}, \mathcal{A}(\cdot), \mu_\phi(\cdot), \mathcal{D}_\psi(\cdot), \epsilon_\theta(\cdot, t), \{\bar{\alpha}_t\}_{t=0}^T, \eta, \delta, \rho, \gamma$, thresholds $t_1 > t_2$

- 1: $\mathbf{z}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 2: **for** $t = T, \dots, 1$ **do**
- 3: $\epsilon_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \hat{\epsilon}_t = \epsilon_\theta(\mathbf{z}_t, t)$
- 4: $\hat{\mathbf{z}}_{0|t} = \frac{1}{\sqrt{\alpha_t}}(\mathbf{z}_t - \sqrt{1 - \alpha_t} \hat{\epsilon}_t)$ ▷ Tweedie's formula
- 5: $\mathbf{z}'_{t-1} = \sqrt{\alpha_{t-1}} \hat{\mathbf{z}}_{0|t} + \sqrt{1 - \alpha_{t-1} - \eta \delta^2} \hat{\epsilon}_t + \eta \delta \epsilon_1$ ▷ DDIM step
- 6: **if** $t > t_1$ **then** ▷ Stage 1: unconditional
- 7: $\mathbf{z}_{t-1} = \mathbf{z}'_{t-1}$
- 8: **else**
- 9: **if** $t > t_2$ and $t \leq t_1$ **then** ▷ Stage 2: image-space optimization
- 10: $\hat{\mathbf{x}}_0 \in \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{y} - \mathcal{A}(\mathbf{x})\|_2^2 + \frac{\rho}{2} \|\mathbf{x} - \mathcal{D}_\psi(\hat{\mathbf{z}}_{0|t})\|_2^2$
- 11: $\mathbf{z}_{0|t}^{\text{dc}} = \mu_\phi(\hat{\mathbf{x}}_0)$ ▷ Re-encode using posterior mean
- 12: **else** ▷ Stage 3: latent-space optimization
- 13: $\mathbf{z}_{0|t}^{\text{dc}} \in \arg \min_{\mathbf{z}} \frac{1}{2} \|\mathbf{y} - \mathcal{A}(\mathcal{D}_\psi(\mathbf{z}))\|_2^2$ ▷ Init. $\hat{\mathbf{z}}_{0|t}$
- 14: **end if**
- 15: $\mathbf{z}_{t-1} = \text{StochasticResample}(\mathbf{z}_{0|t}^{\text{dc}}, \mathbf{z}'_{t-1}, \gamma)$
- 16: **end if**
- 17: **end for**
- 18: **return** $\hat{\mathbf{x}} = \mathcal{D}_\psi(\mathbf{z}_0)$

optimization is stopped at tolerance $\tau = 10^{-4}$. Hard data consistency is applied every 10 reverse steps, and stochastic resampling uses $\gamma = 40$ with the adaptive variance schedule of ReSample.

III. EXPERIMENTS AND RESULTS

We evaluate the proposed framework in three stages. First, we assess the latent AE, since it defines the representation used by the LDM. Second, we evaluate unconditional volume generation and compare the generated distributions with existing 3-D medical generative models. Finally, we study measurement-guided reconstruction against classical baselines and a slice-wise pretrained 2-D DM-prior baseline.

A. Experimental Details

We evaluate the proposed framework on two volumetric medical imaging modalities. For CT, we use 700 volumes from LIDC-IDRI [32] and the CT-LYMPH-NODES collection [33], resampled or cropped to a fixed spatial size of 512^3 . For MRI, we use 540 T1-weighted volumes from the IXI dataset [34], processed to 256^3 .

For CT, intensities are clipped to $[-1000, 1500]$ HU and linearly rescaled to $[-1, 1]$. Volumes with different axial extents are cropped or zero-padded to obtain a fixed 512^3 grid. For MRI, each volume is normalized using the 1st and 99th intensity percentiles, clipped to this range, and rescaled to $[-1, 1]$ before resizing or cropping to 256^3 . All splits are performed at the subject level.

The AE is trained separately for each modality with a downsampling factor $r = 4$, $C_z = 4$ latent channels, and $N = 200,000$ randomly sampled coordinate queries per iteration. This gives latent grids of size 4×128^3 for CT and 4×64^3 for MRI. After training, the AE is frozen and a separate unconditional 3-D LDM is trained for each modality. Both the AE and the 3-D U-Net diffusion model are trained on a single 48 GB GPU, without model parallelism, gradient checkpointing, or patch-wise training.

B. Autoencoding Performance

The AE sets an upper bound on downstream quality, since the diffusion prior operates entirely in latent space and details lost during encoding cannot be recovered during generation or reconstruction. We therefore evaluate the AE first in terms of reconstruction fidelity (peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM)), per-volume runtime, and peak reserved GPU memory.

We compare LIIF-AE with the AEs used in MAISI [9] and 3D MedDiffusion [10]. LIIF-AE encodes the volume once into a compact latent grid and decodes voxel intensities through coordinate queries, with query chunks used only to bound memory. In contrast, MAISI uses overlapping sliding-window autoencoding with window sizes 128^3 and 256^3 , denoted MAISI-W128 and MAISI-W256. For 3D MedDiffusion we evaluate its $4\times$ and $8\times$ patch-volume AEs, decoding the latent with overlapping 128^3 patches. Both baselines blend overlapping predictions using Gaussian-weighted averaging with 50% patch overlap (a stride of half the patch size). Fidelity metrics are reported as mean $\pm$ STD over held-out volumes.

TABLE I: Autoencoding performance on held-out CT and MRI volumes. Time is reported per volume; memory is peak reserved GPU memory.

Dataset	Method	PSNR $\uparrow$	SSIM $\uparrow$	Time $\downarrow$	Memory $\downarrow$
CT 512^3	LIIF-AE	36.81 ± 1.24	0.9725 ± 0.0058	9.55 s	3.27 GB
	MAISI-W128	39.85 $\pm$ 1.69	0.9721 ± 0.0076	301.56 s	3.55 GB
	MAISI-W256	38.32 ± 2.07	0.9749 $\pm$ 0.0041	137.81 s	19.95 GB
	3D MedDiffusion $4\times$	39.76 ± 1.09	0.9300 ± 0.0248	279.72 s	38.86 GB
	3D MedDiffusion $8\times$	37.16 ± 0.46	0.9509 ± 0.0079	115.43 s	7.24 GB
MRI 256^3	LIIF-AE	29.68 ± 0.52	0.9022 ± 0.0165	2.95 s	1.21 GB
	MAISI-W128	31.80 $\pm$ 0.78	0.9361 $\pm$ 0.0124	22.68 s	2.68 GB
	MAISI-W256	30.96 ± 0.75	0.9293 ± 0.0140	5.53 s	18.49 GB
	3D MedDiffusion $4\times$	30.79 ± 0.71	0.9241 ± 0.0138	13.59 s	6.25 GB
	3D MedDiffusion $8\times$	25.29 ± 0.54	0.7418 ± 0.0325	11.12 s	6.36 GB

Across both modalities (Table I), the strongest baseline (MAISI) leads on fidelity by a small margin, while LIIF-AE is decisive on cost, achieving the lowest runtime and the lowest memory in every row. On CT, LIIF-AE matches the best SSIM (0.9725 vs. 0.9749) and trails the best PSNR by roughly 3 dB, yet autoencodes a full 512^3 volume in 9.55 s at 3.27 GB. This is more than an order of magnitude faster than every baseline, which ranges from 115 to 302 s.

The cost of the baselines follows from their decoding strategy. A dense full-resolution decode is infeasible under a practical memory budget, so the volume is decoded in overlapping sub-volumes, the full 3-D decoder is re-evaluated on each, and the outputs are blended. Runtime therefore scales with the number of sub-volumes, forcing a trade-off between memory and speed that neither baseline escapes. 3D MedDiffusion at $4\times$ reaches competitive fidelity only at 38.86 GB, whereas MAISI-W128 contains memory but inflates runtime to 301.56 s by multiplying the forward passes. LIIF-AE avoids this regime by running the convolutional decode once on the compact latent and rendering the volume through coordinate queries. The chunks split the query list rather than the volume and require no stitching. The qualitative MRI and CT comparisons in Figs. 1 and 2 confirm that this single latent decode preserves anatomy across the axial, coronal,

and sagittal views, with no visible patch seams relative to the ground truth.

The MRI results follow the same pattern. MAISI-W128 leads on fidelity (PSNR 31.80, SSIM 0.9361) and LIIF-AE trails modestly (PSNR 29.68, SSIM 0.9022), which we attribute to mild smoothing of fine detail by the compact latent. This is consistent with the qualitative comparison in Fig. 1, where the small fidelity gap appears only as slightly softer high-frequency structure rather than as structured artifacts. LIIF-AE decodes a 256^3 volume in 2.95 s at 1.21 GB, roughly $8\times$ faster and at less than half the memory of MAISI-W128.

LIIF-AE thus trades a small and bounded loss in voxel-wise fidelity for large and consistent reductions in runtime and memory. Across both modalities (Figs. 1 and 2), the volume is rendered continuously through coordinate queries on a single latent decode, so the reconstruction is free of patch boundaries and stitching artifacts by construction. This efficiency is not merely a convenience. In guided reconstruction, the decoder is embedded in an iterative loop and is evaluated many times per volume as the prior is combined with the measurement model. Its per-call cost therefore multiplies across iterations and dominates the total reconstruction time. A decoder that autoencodes a volume in seconds rather than minutes, with no patch seams to propagate through the iterations, is what makes reconstruction with this prior practical (Section III-D).

C. Unconditional Volume Generation

We assess unconditional generation with a 2.5-D Fréchet Inception Distance (FID) protocol. Each volume is sliced along the axial (XY), coronal (YZ), and sagittal (ZX) planes; per-slice features are extracted with a RadImageNet-pretrained ResNet-50 [35], and FID is computed per plane and averaged. Preprocessing is identical across methods, and we use 40 unseen volumes per dataset from LIDC-IDRI CT and IXI MRI T1.

For MRI, IXI is anonymized with a face mask that we could apply to our samples but not to the MAISI samples, so the reference and the two methods do not share a common spatial support. We therefore report the MRI scores as indicative only and draw no ranking from them. 3D MedDiffusion is excluded from MRI altogether, as its generator synthesizes only the brain parenchyma and omits the skull and surrounding anatomy present in IXI, precluding a meaningful whole-head comparison.

TABLE II: Unconditional generation quality.

Dataset	Method	FID _{XY}	FID _{YZ}	FID _{ZX}	FID _{avg}
CT	Ours	3.955	5.726	4.646	4.776
	MAISI	3.788	4.751	3.947	4.162
	3D MedDiffusion	2.375	4.682	3.253	3.437
MRI	Ours	4.379	4.740	4.602	4.574
	MAISI	5.769	6.047	6.555	6.124

On the matched-support CT benchmark (Table II), 3D MedDiffusion attains the lowest FID, followed by MAISI and our method. Our scores remain within the range of these diffusion baselines while using only a continuous coordinate decoder and the substantially lower decoding cost of Section III-B. On

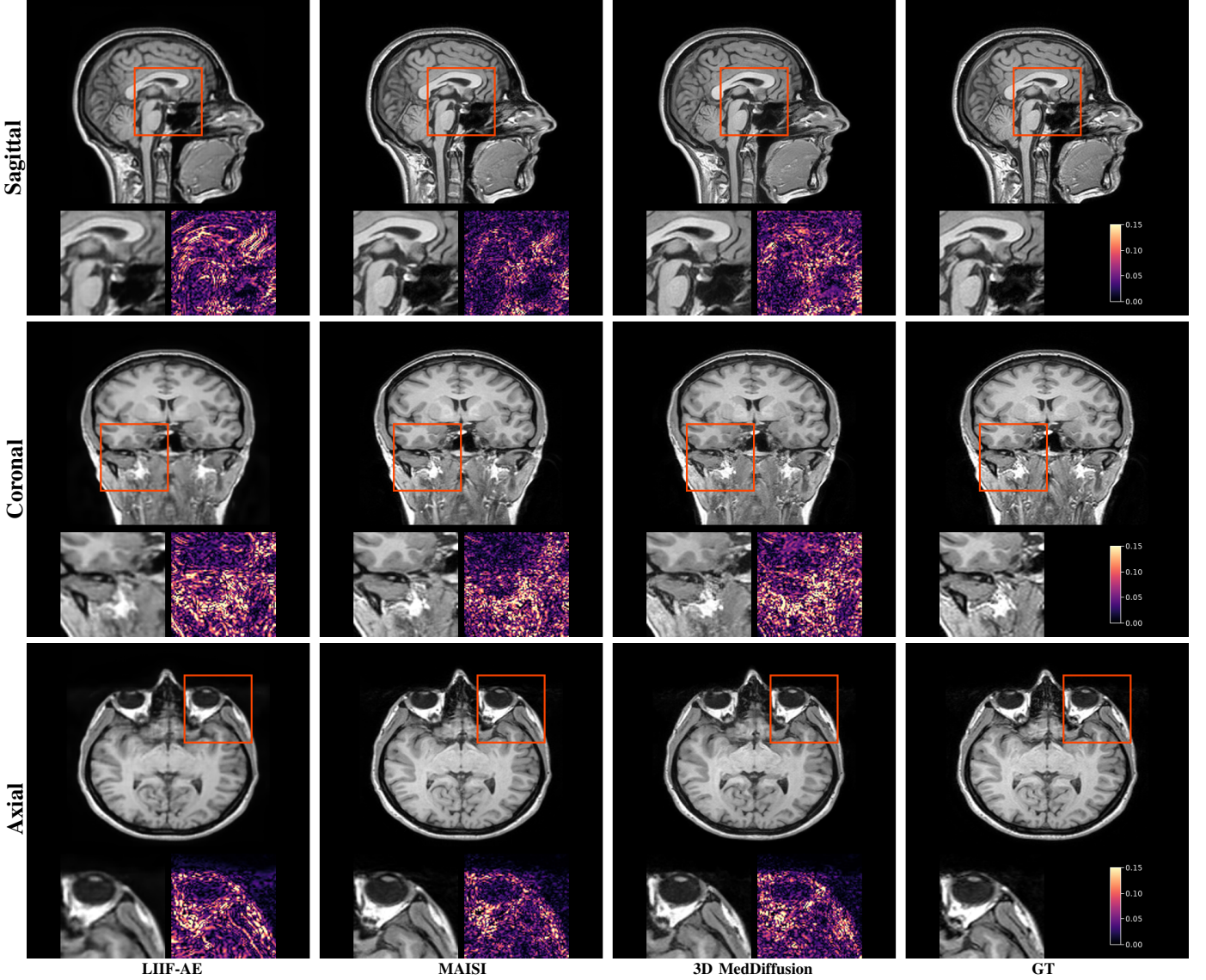


Fig. 1: Qualitative comparison of MRI autoencoding on a representative IXI volume. Rows show central sagittal, coronal, and axial slices. For each method, the top panel shows the decoded slice and selected region of interest, while the bottom panels show its magnification and absolute error map. Columns compare LIIF-AE, MAISI, 3D MedDiffusion, and the input volume.

IXI MRI, our method reports lower FID on every plane and on average (4.574 vs. 6.124); we attribute part of this gap to the unmatched masking support rather than to sample quality, and treat it as indicative only. The qualitative comparison in Fig. 3 is consistent with these scores: all methods produce anatomically coherent volumes across the three views, with our samples showing no patch seams or slice discontinuities. Overall, our compact representation generates full volumes of quality comparable to strong diffusion baselines at a fraction of their decoding cost.

D. Image Reconstruction

We evaluate the frozen latent prior $p(z)$ as a regularizer for the inverse problem (4) on sparse-view CT and accelerated single-coil MRI. Data consistency is imposed through $\mathcal{A} \circ \mathcal{D}_\psi$, with coordinate queries processed in memory-bounded chunks.

For CT, the parallel-beam projection operator is implemented using TorchRadon [36], with 30 or 60 noisy views. For MRI, we use $\mathcal{A}(x) = M\mathcal{F}(x)$, where $\mathcal{F}$ is the slice-wise two-dimensional Fourier transform and M is a variable-density Cartesian mask. The central 8% of k -space is fully sampled, while the remaining lines are undersampled to obtain acceleration factors $R = 4$ and $R = 8$.

Among approaches that extend pretrained 2-D diffusion priors to volumetric reconstruction, DiffusionMBIR introduces inter-slice regularization [37], whereas TPDM combines priors trained on perpendicular planes [38]. We use decomposed diffusion sampler (DDS) [39] as a strong representative of this family because it couples slice-wise denoising with conjugate-gradient data consistency and has demonstrated improved reconstruction quality and lower sampling cost than earlier volumetric diffusion solvers.

Accordingly, the quantitative comparison is made against

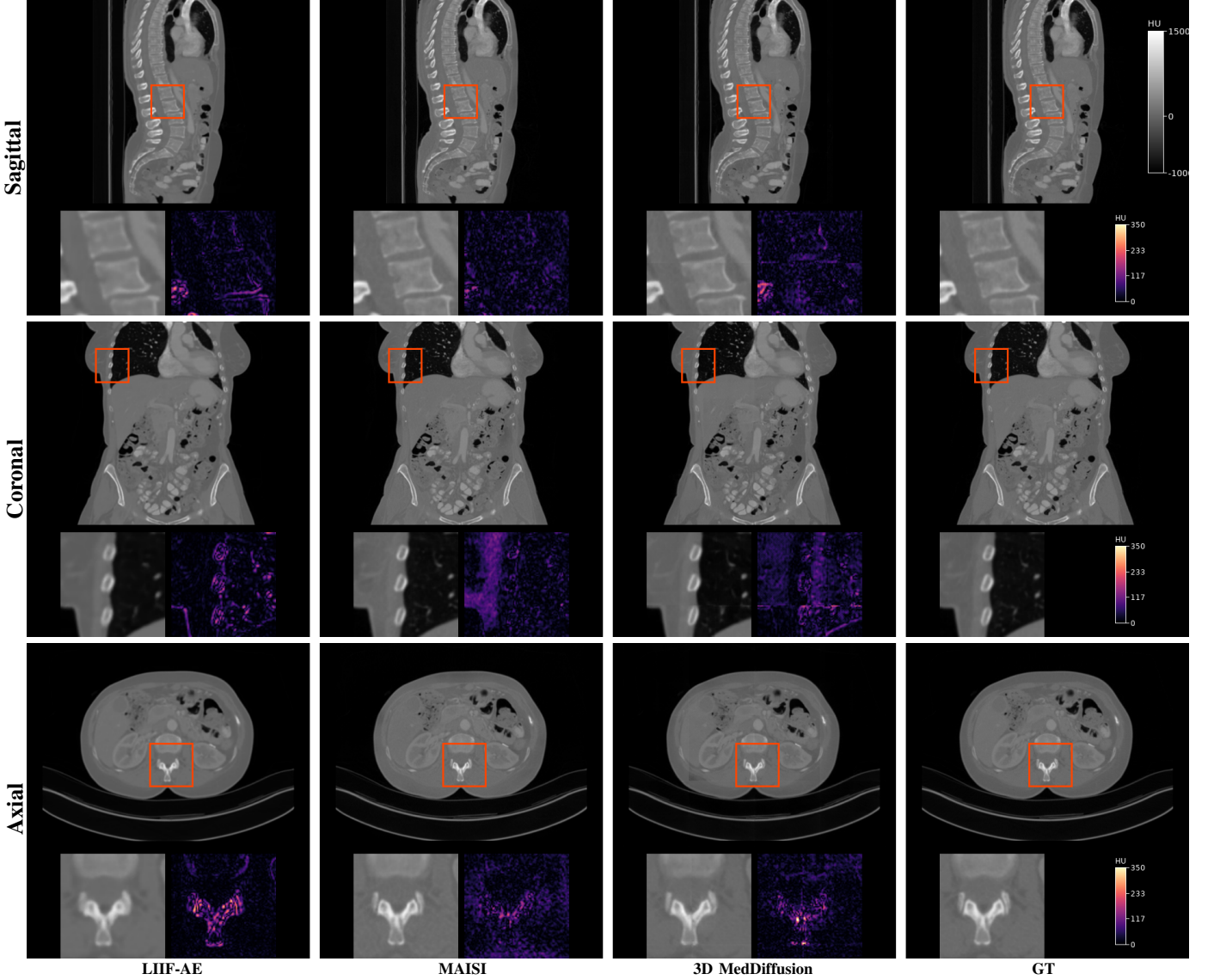


Fig. 2: Qualitative comparison of CT autoencoding on a representative held-out volume. Rows show central sagittal, coronal, and axial slices. For each method, the top panel shows the decoded slice and selected region of interest, while the bottom panels show its magnification and absolute error map in HU. Columns compare LIIF-AE, MAISI, 3D MedDiffusion, and the input volume. Decoder outputs are converted to HU using the range $[-1000, 1500]$.

DDS, while filtered back-projection (FBP) and zero-filled (ZF)-inverse fast Fourier transform (IFFT) are included as analytic and zero-filled references for CT and MRI, respectively. Results are evaluated in HU for CT and as magnitude images for MRI. PSNR and SSIM are computed against the ground truth and averaged over axial, coronal, and sagittal slices.

The latent prior reconstructs high-quality volumes from severely undersampled measurements (Table III). On CT it reaches 34.20 dB PSNR at 30 views and 39.31 dB at 60 views; on MRI it reaches 29.03 dB at $\times 4$ and 27.12 dB at $\times 8$, with corresponding gains in SSIM over the zero-filled reference. DDS scores highest in every setting, as expected, since it runs a full-resolution diffusion model on the volume itself while our prior works in the compact latent space and reconstructs through the decoder. Our method stays close to DDS on CT, within about 4 dB PSNR at 60 views, and the qualitative

TABLE III: Reconstruction quality from undersampled measurements: sparse-view CT (30 and 60 projections) and accelerated MRI ($\times 4$, $\times 8$). PSNR and SSIM are averaged over axial, coronal, and sagittal 2-D slices (higher is better).

Modality	Setting	Method	PSNR	SSIM
CT	30 views	FBP	10.18	0.1268
		DDS	42.17	0.9541
		Ours	34.20	0.9075
	60 views	FBP	12.03	0.1785
		DDS	43.50	0.9548
		Ours	39.31	0.9514
MRI	$\times 4$	ZF	25.02	0.6610
		DDS	36.83	0.9628
		Ours	29.03	0.8901
	$\times 8$	ZF	22.52	0.6218
		DDS	33.22	0.9282
		Ours	27.12	0.8401

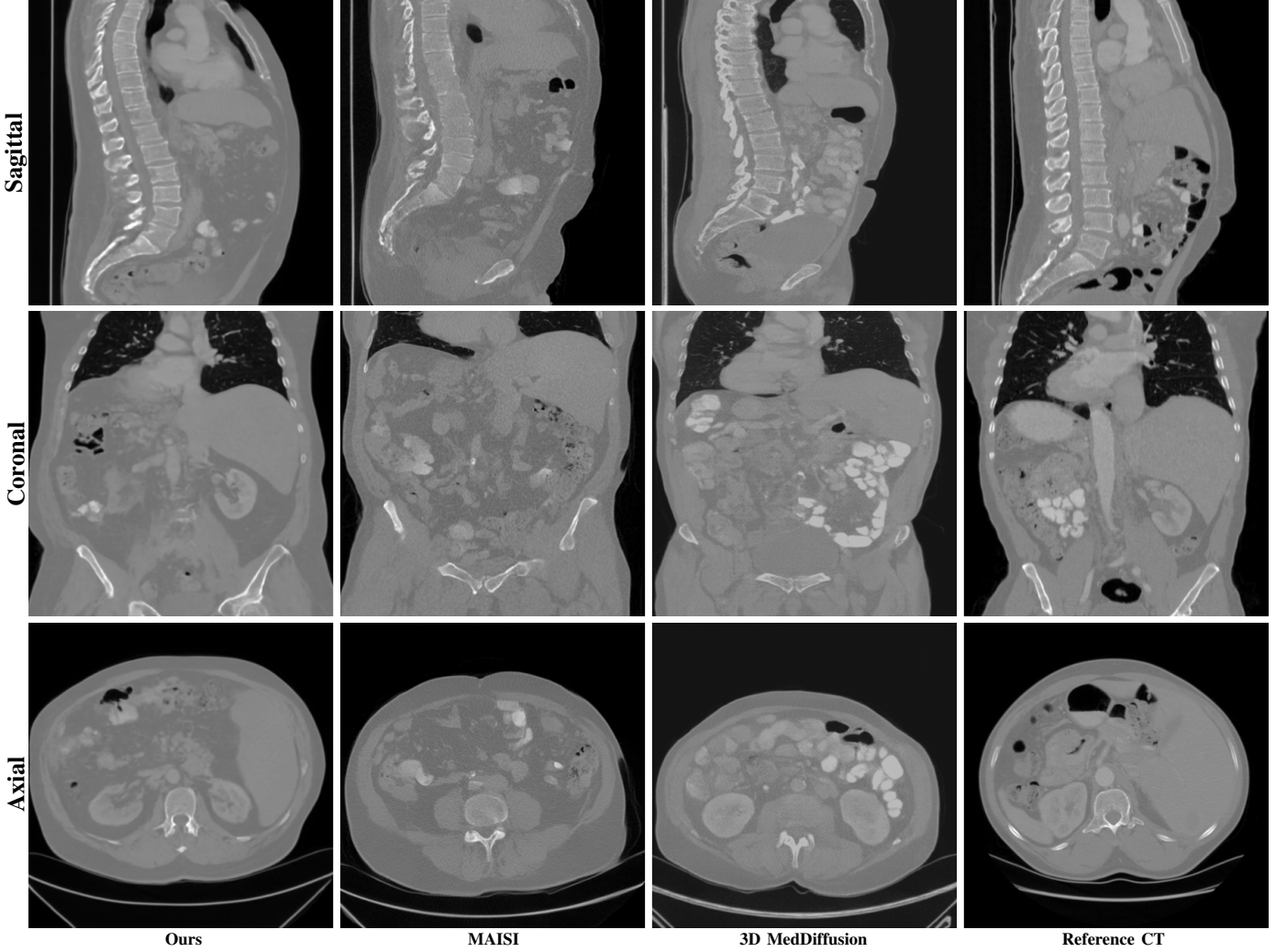


Fig. 3: Qualitative comparison of unconditional CT volume generation. Rows show central sagittal, coronal, and axial slices. Columns compare samples generated by the proposed method, MAISI, and 3D MedDiffusion, together with an unpaired real CT volume shown for reference. All volumes are displayed in HU using the same window.

comparisons (Figs. 4–7) show sharp anatomy without the streak and aliasing artifacts of the analytic reconstructions. The distinction from DDS is the prior itself: DDS guides reconstruction with a 2-D slice-wise DM, whereas ours is a native 3-D generative prior applied through the same decoder used for generation.

IV. DISCUSSION

The results place the main contribution of this work in the AE, rather than in the diffusion process. For matched latent resolution, denoiser architecture, precision, and sampling schedule, the diffusion-stage cost is comparable to that of a conventional 3-D LDM. The proposed AE instead changes the cost of representing and decoding high-resolution volumes. In our single-GPU setting, it provides a common volumetric representation that can be used for both generation and reconstruction, without introducing a cheaper diffusion sampler.

The autoencoding experiments show that the decoding advantage follows from where computation is repeated. Patch-

based approaches execute the complete decoder for every sub-volume, recomputing intermediate activations and, when patches overlap, blending redundant predictions. Our decoder evaluates the convolutional feature stage once per decoder call on the latent grid and reuses the resulting features across all output locations. Only feature lookup and the lightweight implicit head are repeated over memory-bounded coordinate batches. The measured reductions in decoding time and memory therefore result from avoiding repeated full-decoder passes, which is particularly relevant when reconstruction requires many decoder evaluations.

This factorization also establishes the main limitation of the method. The decoder preserves large-scale anatomy but smooths some fine structures and high-frequency texture. Although each prediction uses locally aggregated latent features, the training objective provides no explicit supervision over coherent decoded neighborhoods. Perceptual or adversarial objectives could strengthen local detail, but would require rendering structured regions and increase training memory. Because the AE is frozen before diffusion training, its representational

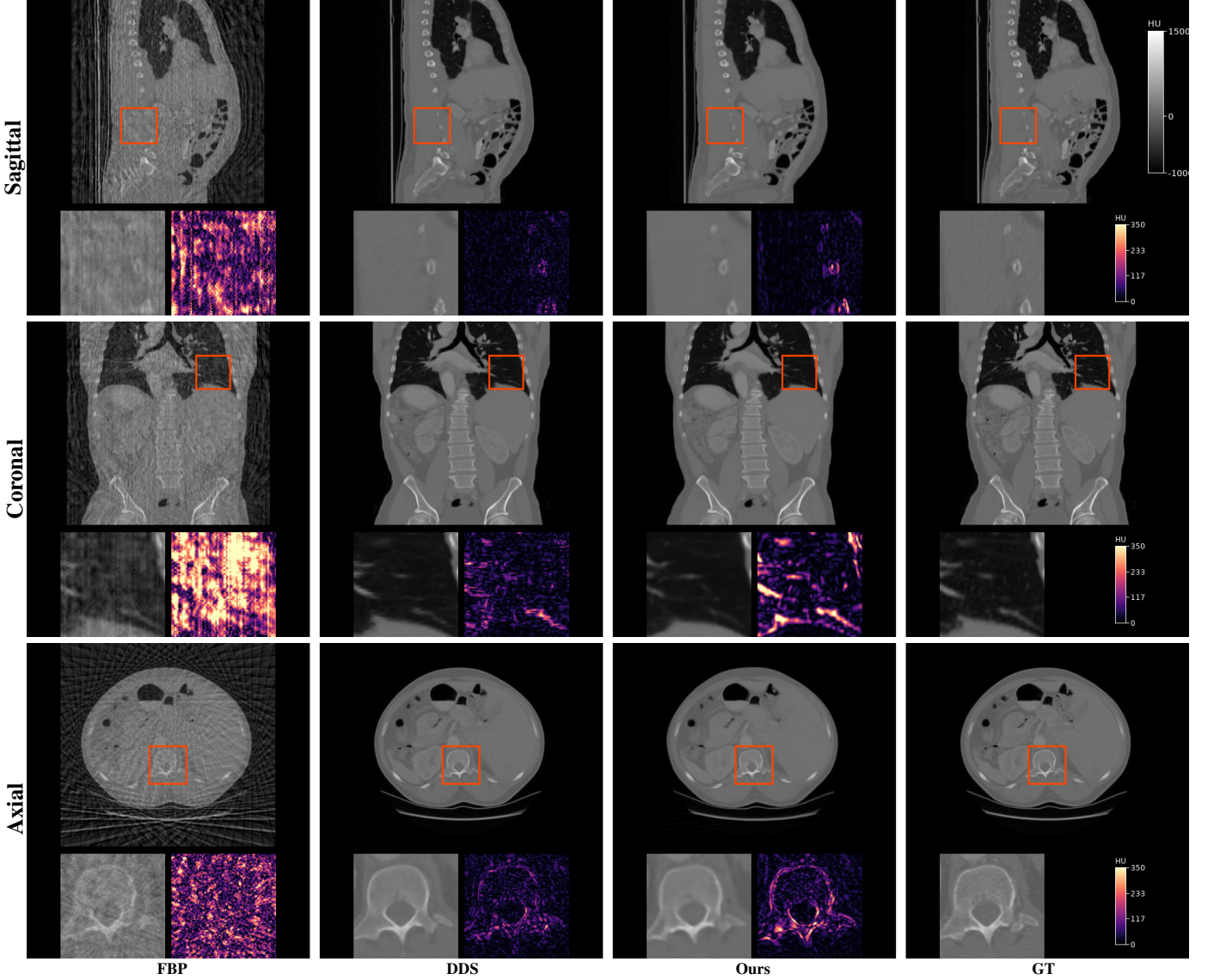


Fig. 4: Qualitative comparison of sparse-view CT reconstruction from 30 noisy projections. Rows show sagittal, coronal, and axial slices. For each method, the top panel shows the reconstruction and selected region of interest, while the bottom panels show its magnification and absolute error map in HU. Columns compare FBP, DDS, our method, and the ground truth. All images use the same HU window.

range sets a fidelity ceiling: information not preserved during autoencoding cannot be recovered through improved latent denoising. Generation and reconstruction therefore inherit the same limitation.

For generation, the method produces qualitatively coherent full volumes without visible patch seams, but its CT FID remains behind the strongest baselines. This gap likely reflects both the limited scale and diversity of the modality-specific training cohorts and the smoothing introduced by the frozen AE, which is inherited by all generated samples and affects the feature statistics measured by FID. The results therefore demonstrate efficient full-volume synthesis rather than state-of-the-art distribution matching. The MRI scores should be interpreted cautiously because the face-masked IXI references and unmasked MAISI samples do not share the same spa-

tial support, while 3D MedDiffusion is excluded because it generates only brain parenchyma. Since voxel- or wavelet-domain priors may remain tractable at 256^3 resolution [40], the decoder-level benefit is clearer in the 512^3 CT setting.

The reconstruction results expose the autoencoder limitation more strongly. DDS [39] achieves higher PSNR and SSIM in every evaluated setting because its data-consistency updates act directly on image pixels, without passing through an autoencoder, while its 2-D denoiser benefits from a large number of slice-level training instances. Combined with axial total variation (TV) regularization, this yields a strong volumetric reconstruction baseline [41]. In contrast, our method optimizes through the composite mapping $\mathcal{A} \circ \mathcal{D}_\psi$, which is nonlinear and generally nonconvex even when $\mathcal{A}$ is linear [29], [42]. Image-space corrections must be mapped back

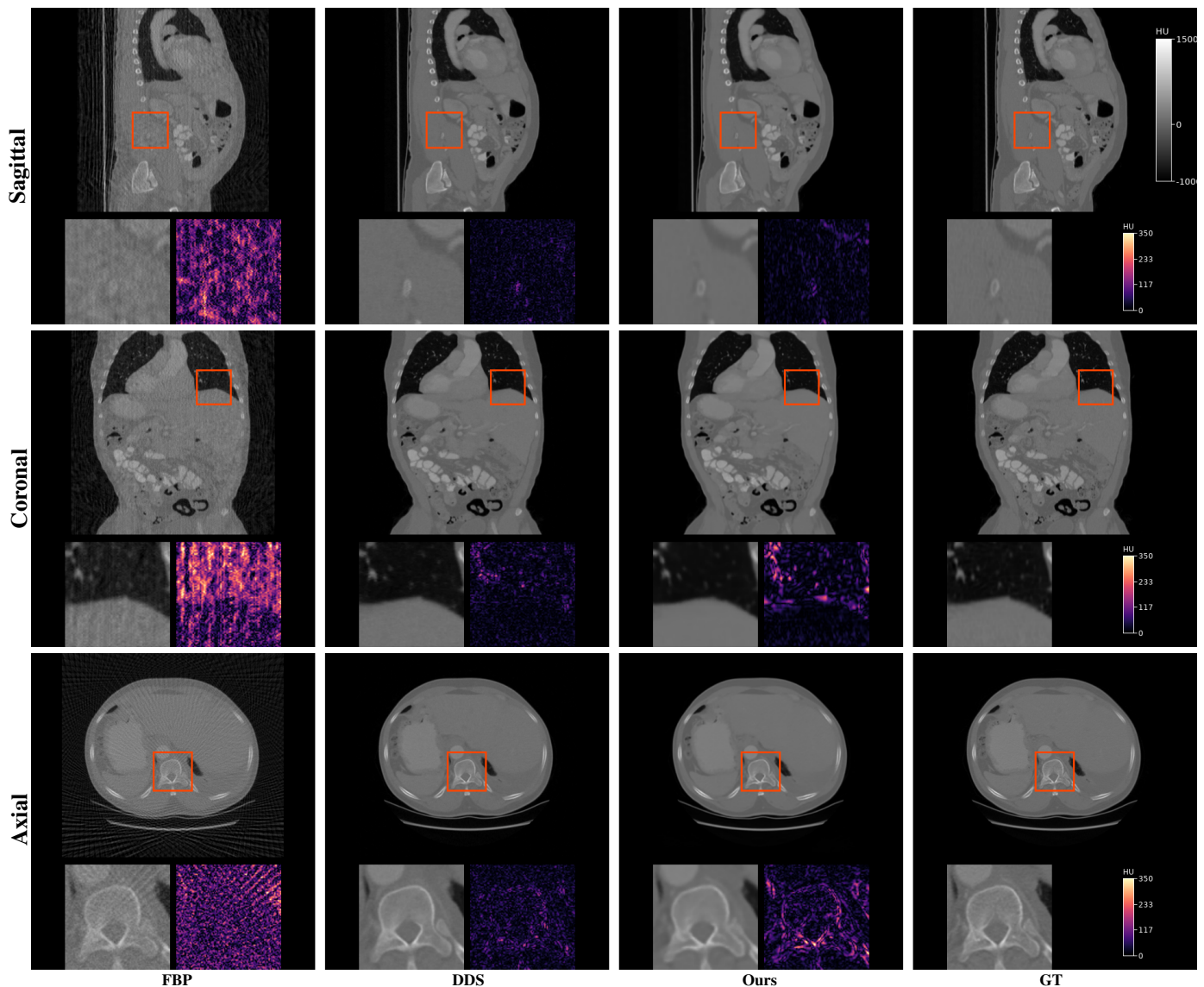


Fig. 5: Qualitative comparison of sparse-view CT reconstruction from 60 noisy projections. Rows show sagittal, coronal, and axial slices. For each method, the top panel shows the reconstruction and selected region of interest, while the bottom panels show its magnification and absolute error map in HU. Columns compare FBP, DDS, our method, and the ground truth. All images use the same HU window.

through an approximately invertible and lossy encoder, which can attenuate recovered details [29], [31], whereas latent-space corrections avoid this projection at the cost of repeated differentiation through the decoder and remain restricted to signals representable by the frozen AE. This combination of indirect data consistency, autoencoder loss, and constrained latent optimization accounts for much of the quantitative gap to DDS. The reconstruction experiments should therefore be interpreted as a demonstration that a frozen 3-D latent prior trained for generation can also be incorporated into a hard-data-consistency procedure for sparse-view CT and accelerated MRI, rather than as evidence of superiority over a regularized 2-D pixel-domain prior.

Taken together, the results establish a trade-off between decoder efficiency and voxel-level fidelity. The proposed fac-

torization is most useful when repeated dense volumetric decoding is the dominant computational constraint, but its benefit diminishes when image-space processing remains tractable or fine-detail preservation is the primary objective. The conclusions are also limited by the use of two public datasets, simulated sparse-view CT, and simulated single-coil MRI. The study does not include external clinical validation, multi-coil acquisitions, realistic acquisition imperfections, reader studies, or explicit volumetric-consistency metrics.

Future work should address both decoder fidelity and diffusion cost. A convolutional local image function (CLIF)-style decoder could use broader local feature context and neighborhood-aware supervision to recover high-frequency detail while preserving memory-bounded coordinate evaluation [43], whereas bit-plane decomposition offers a comple-

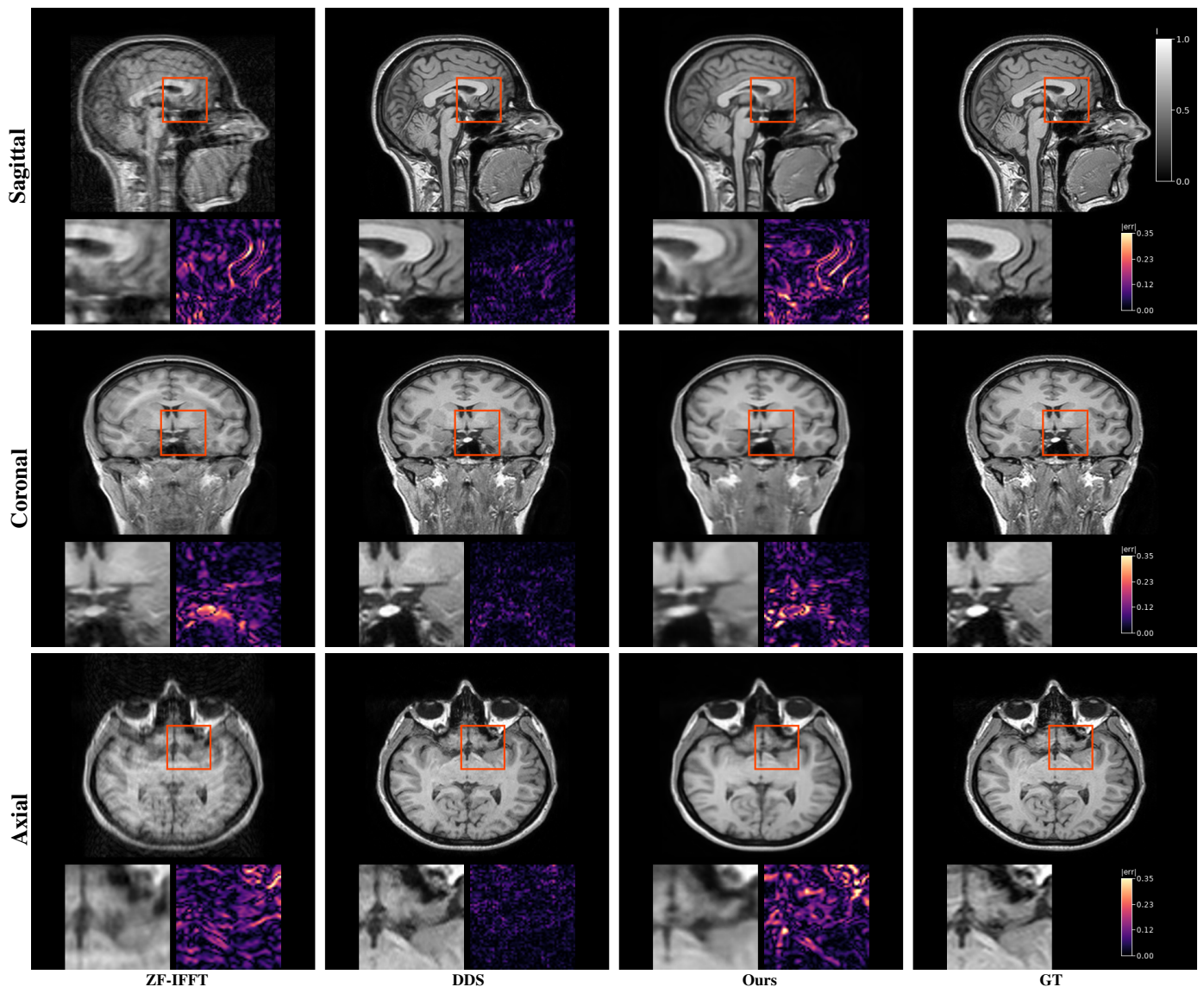


Fig. 6: Qualitative comparison of single-coil Cartesian MRI reconstruction at $R = 4$. Rows show sagittal, coronal, and axial slices. For each method, the top panel shows the reconstruction and selected region of interest, while the bottom panels show its magnification and absolute error map. Columns compare ZF-IFFT, DDS, our method, and the ground truth. All images use the same normalized intensity window.

mentary route toward near-lossless implicit representations by predicting binary bit-planes instead of a single continuous intensity [44]. Adapting these ideas to a latent-conditioned 3-D medical decoder could raise the autoencoding fidelity ceiling without returning to repeated dense decoding. At the same time, more compact latent representations should be explored to reduce the cost of diffusion itself. Autoencoders with spatial downsampling factors of up to 32 substantially reduce the number of latent tokens processed during denoising [45], [46], suggesting that compact or anisotropic 3-D latent grids could lower both training and sampling costs for high-resolution medical volumes. The central challenge is to obtain this additional compression without further degrading fine structures or widening the reconstruction gap, while larger and more heterogeneous training cohorts and more efficient latent data-consistency methods remain complementary directions.

V. CONCLUSION

We introduced a continuous 3-D LDM for high-resolution medical volume generation and reconstruction. The proposed autoencoder combines a compact volumetric latent representation with a coordinate-conditioned implicit decoder, allowing full CT and MRI volumes to be rendered without patch-wise decoding or fixed output grids. This design enables 512^3 CT generation on a single GPU and produces qualitatively coherent volumes without visible patch seams.

The same frozen 3-D diffusion prior was also applied to sparse-view CT and accelerated MRI through hard data consistency, without task-specific retraining. Although the lossy autoencoder limits fine-detail recovery and leads to lower reconstruction accuracy than direct pixel-domain methods such as DDS, the results demonstrate that one volumetric

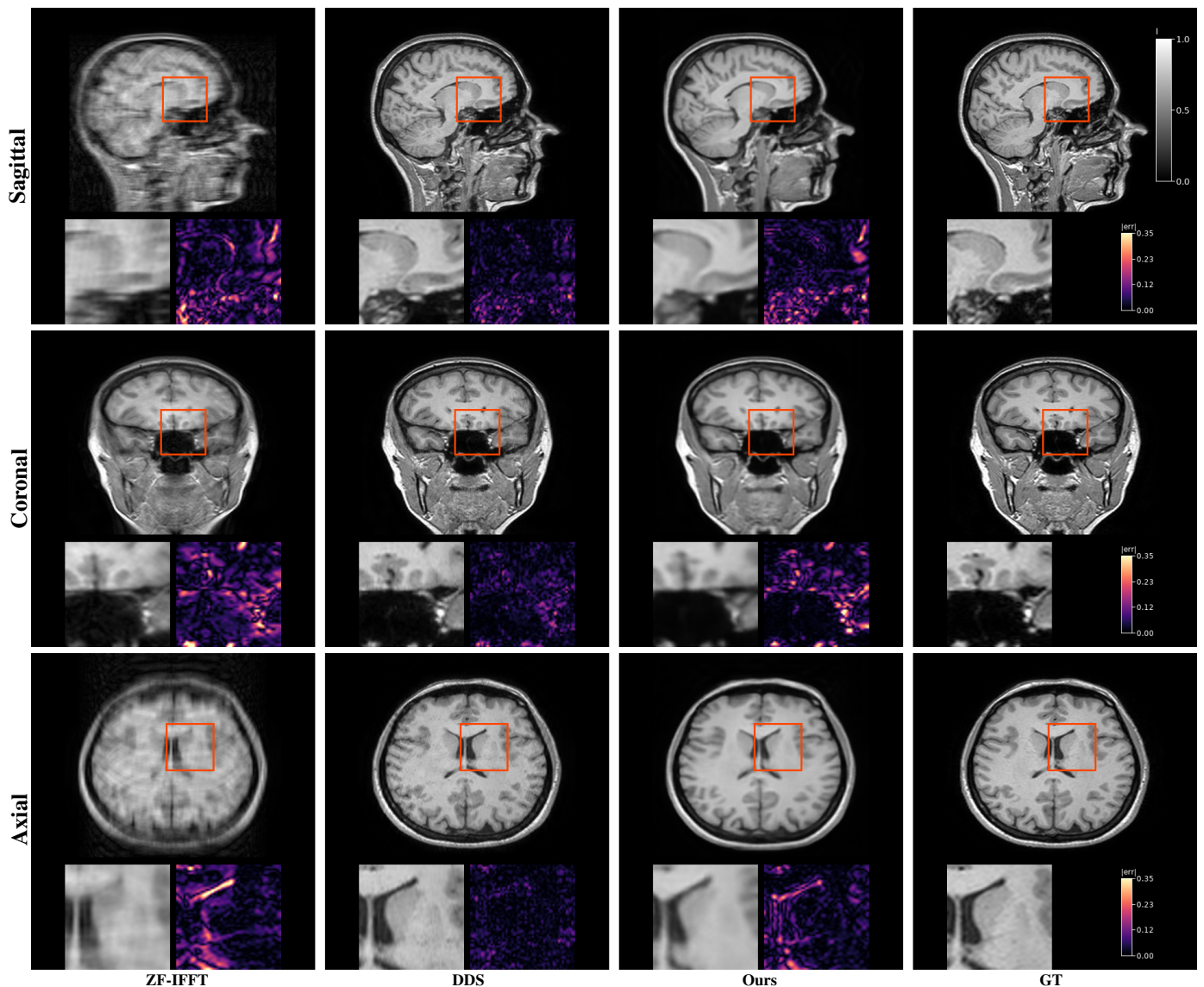


Fig. 7: Qualitative comparison of single-coil Cartesian MRI reconstruction at $R = 8$. Rows show sagittal, coronal, and axial slices. For each method, the top panel shows the reconstruction and selected region of interest, while the bottom panels show its magnification and absolute error map. Columns compare ZF-IFFT, DDS, our method, and the ground truth. All images use the same normalized intensity window.

latent prior can support both unconditional generation and inverse-problem reconstruction. Overall, the framework offers a practical route to continuous high-resolution medical volume modeling, with decoder fidelity remaining the main target for further improvement.

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